EELU
الجامعة المصرية للتعلم الإلكتروني الأهلية
THE EGYPTIAN E-LEARNING UNIVERSITY

Fall Semester

2025-2026

Information Technology

IT438

Communication Technology

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Acknowledgement: *This presentation contains some figures and text from Data Communications and Networks book by W. Stallings*

Optical Modulation

Optical Modulation

Optical modulation is specific to systems that use light waves (usually from lasers or LEDs) as the carrier. It involves altering the properties of light (amplitude, phase, frequency, polarization) to carry information over optical fibers.

Common Digital Modulation Techniques in Optical Fiber

1. Amplitude Shift Keying (ASK) / Intensity Modulation (IM)
2. Phase Shift Keying (PSK)
3. Quadrature Amplitude Modulation (QAM)
4. Frequency Shift Keying (FSK)

Optical Modulation

1- Amplitude Shift Keying (ASK) / Intensity Modulation (IM)

ASK, often referred to as Intensity Modulation (IM) in optical communication, is the simplest and most widely used technique in optical fiber systems. In IM, the intensity of the light wave (usually from a laser or LED) is modulated to carry data.

- Binary 1: High intensity (light "on").
- Binary 0: Low intensity (light "off").

2- Phase Shift Keying (PSK)

PSK, particularly for long-distance and high-speed transmission. Since PSK uses phase changes to represent data, it can provide more reliable communication over noisy and long optical links.

- Binary Phase Shift Keying (BPSK): Encodes data using two phases (0° and 180°).
- Quadrature Phase Shift Keying (QPSK): Encodes data using four phases (0°, 90°, 180°, 270°), allowing two bits per symbol.

Optical Modulation

3- Quadrature Amplitude Modulation (QAM)

QAM is increasingly used in high-speed optical communication systems due to its efficiency in transmitting large amounts of data over a given bandwidth. QAM modulates both the amplitude and phase of the light wave to encode multiple bits per symbol.

- 16-QAM: Transmits 4 bits per symbol (16 amplitude/phase combinations).
- 64-QAM: Transmits 6 bits per symbol (64 amplitude/phase combinations).

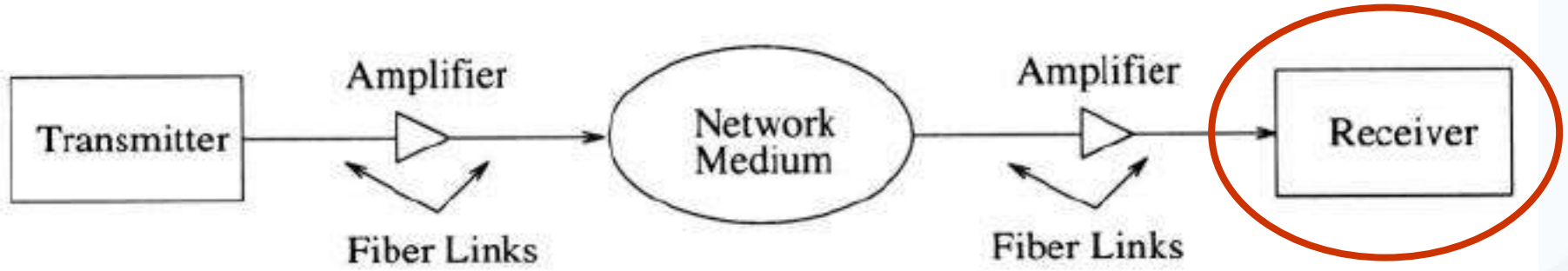
4- Frequency Shift Keying (FSK)

FSK can be used in optical systems, but it is *not as common as IM or PSK* due to its lower spectral efficiency. Minimum Shift Keying (MSK), a variation of FSK, has been implemented in some optical systems for its bandwidth efficiency and lower noise sensitivity.

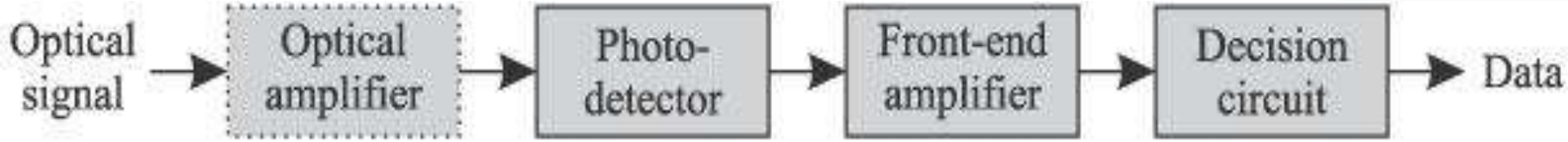
Summary of Digital Modulation in Optical Fiber

Modulation Method	Used in Optical Fiber?	Comments
ASK (IM)	Yes	Simple, widely used in short-distance systems (e.g., OOK).
PSK	Yes	Common in long-distance/high-speed systems (QPSK, BPSK, DPSK).
QAM	Yes	Highly efficient, used in high-speed coherent systems (e.g., 16-QAM, 64-QAM).
FSK	Rarely	Less common, sometimes used in niche applications (e.g., MSK).

Receivers/Filters



Optical Receivers: Basic Idea



- Optical receiver is responsible for converting the received optical signal (light) back into an electrical signal. It consists of several components that work together to ensure the signal is properly detected, amplified, and decoded.
- Two types of optical receivers:
  1. Direct detection
  2. Coherent detection

Optical Receivers

1. Direct Detection Receiver

- It converts the optical signal directly into an electrical signal using a photodetector (e.g., a photodiode).
- The optical signal is incident on a photodetector which converts the incoming light into a corresponding electrical current (proportional to the intensity of light).
- The electrical signal is then amplified and processed to recover the transmitted data.
- Used in low- and medium-speed optical communication systems like fiber-to-the-home (FTTH), data centers, and short-range communication.

Optical Receivers

1. Direct Detection Receiver (cont.)

- Advantages:
  - Simple and cost-effective.
  - Suitable for Intensity Modulation/Direct Detection (IM/DD) systems, like On-Off Keying (OOK).
- Limitations:
  - Limited performance in terms of sensitivity and data rate compared to coherent detection.
  - Exposed to noise and dispersion for long-distance communication.

Optical Receivers

2. Coherent Detection Receiver

- It is an advanced type of optical receiver that uses both the amplitude and phase of the optical signal, allowing for more complex modulation formats and higher sensitivity.
- The optical signal is combined with a local oscillator (a reference laser) using a device called a hybrid, then detected by the photodetector.
- By comparing the phase and amplitude of the received signal to the local oscillator's reference, the receiver can detect not just the intensity but also the phase and frequency shifts of the signal.
- Used in long-haul optical networks, and high-speed optical systems.

Optical Receivers

2. Coherent Detection Receiver (cont.)

- Advantages:
  - Higher sensitivity than direct detection.
  - Supports advanced modulation formats like QPSK and QAM, increasing data rates and spectral efficiency.
  - Can compensate for impairments like chromatic dispersion and polarization mode dispersion (PMD) using digital signal processing (DSP).
- Limitations:
  - More complex and expensive than direct detection.
  - Requires precise synchronization with the local oscillator, making the receiver design more challenging.

Optical Filters

- Used to selectively transmit or block certain wavelengths of light while allowing others to pass
- In **multi-channel systems**, requires careful control over which wavelengths are allowed to reach the receiver and which should be filtered out. This is where optical filters come in.
- **Key Characteristics of Optical Filters**
  1. Center Wavelength
  2. Bandwidth
  3. Rejection Band
  4. Insertion Loss
  5. Out-of-Band Rejection

Optical Filters (cont.)

Types of Optical Filters:

Optical filters are classified based on their function, design, and the type of filtering mechanism they use.

1- Bandpass Filters

It allows light of a certain wavelength range (the passband) to pass through, while rejecting or attenuating wavelengths outside this range (the stopband).

2- Fiber Bragg Gratings (FBG)

- It is a periodic variation in the refractive index of the fiber core, which reflects specific wavelengths and transmits others.

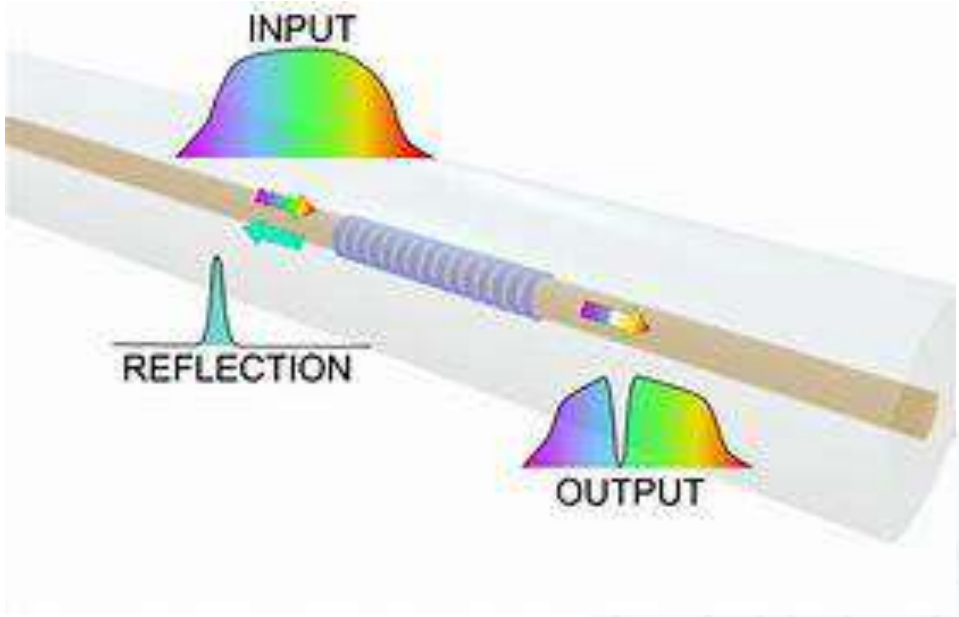
3- Tunable Optical Filters

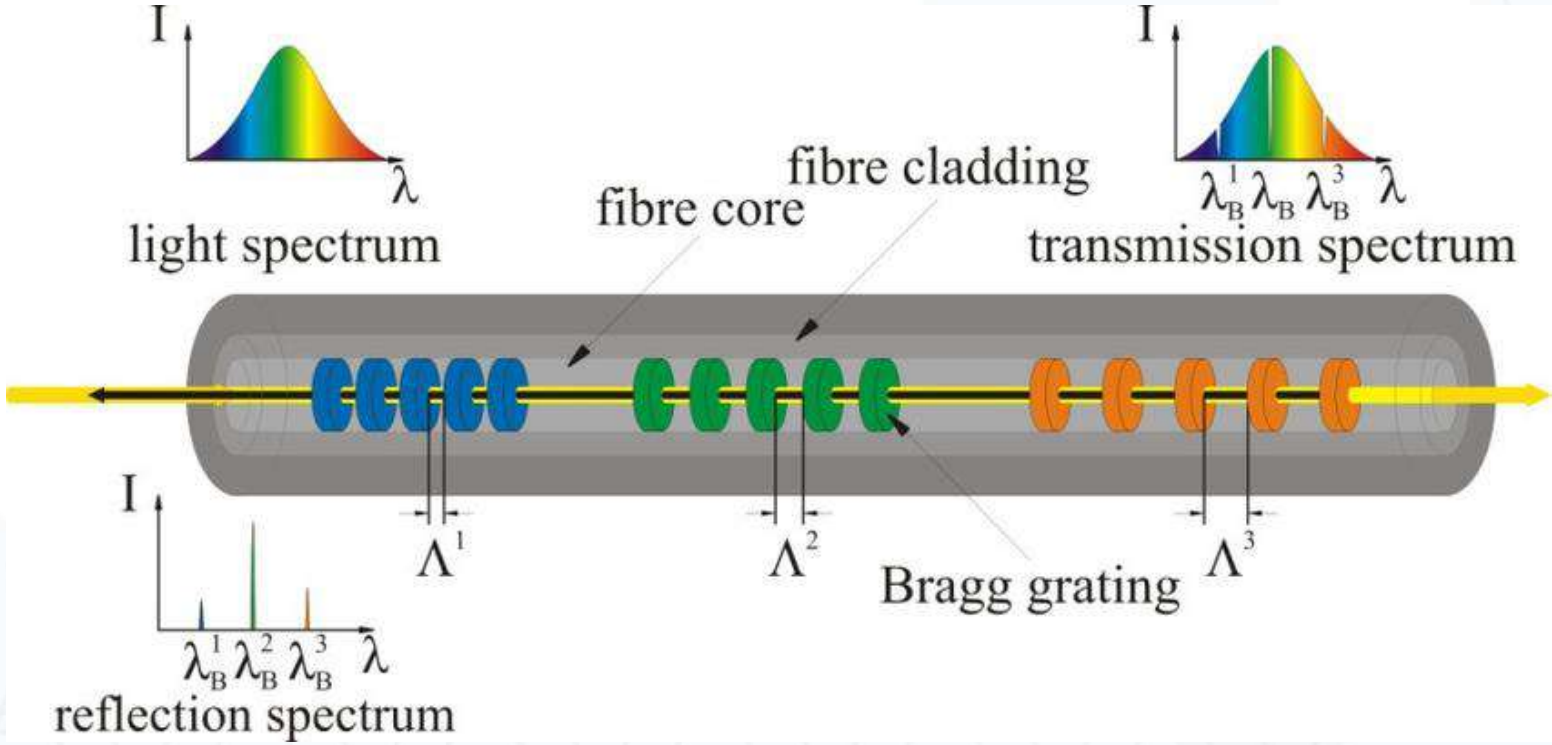
- It allows the user to adjust the center wavelength of the passband or stopband, making them highly flexible for dynamic optical systems.

Optical Filters (cont.)

Filter Type	Typical WDM Application	Advantages
Bandpass Filters	Used to isolate and pass specific wavelength channels	High selectivity, commonly used in multiplexing/demultiplexing
Fiber Bragg Gratings (FBG)	Reflects specific wavelengths, used for channel separation and dispersion compensation	Compact, effective for channel filtering and management
Tunable Optical Filters	ROADMs, dynamic WDM systems	Flexibility to tune different channels, dynamic operation

Gratings

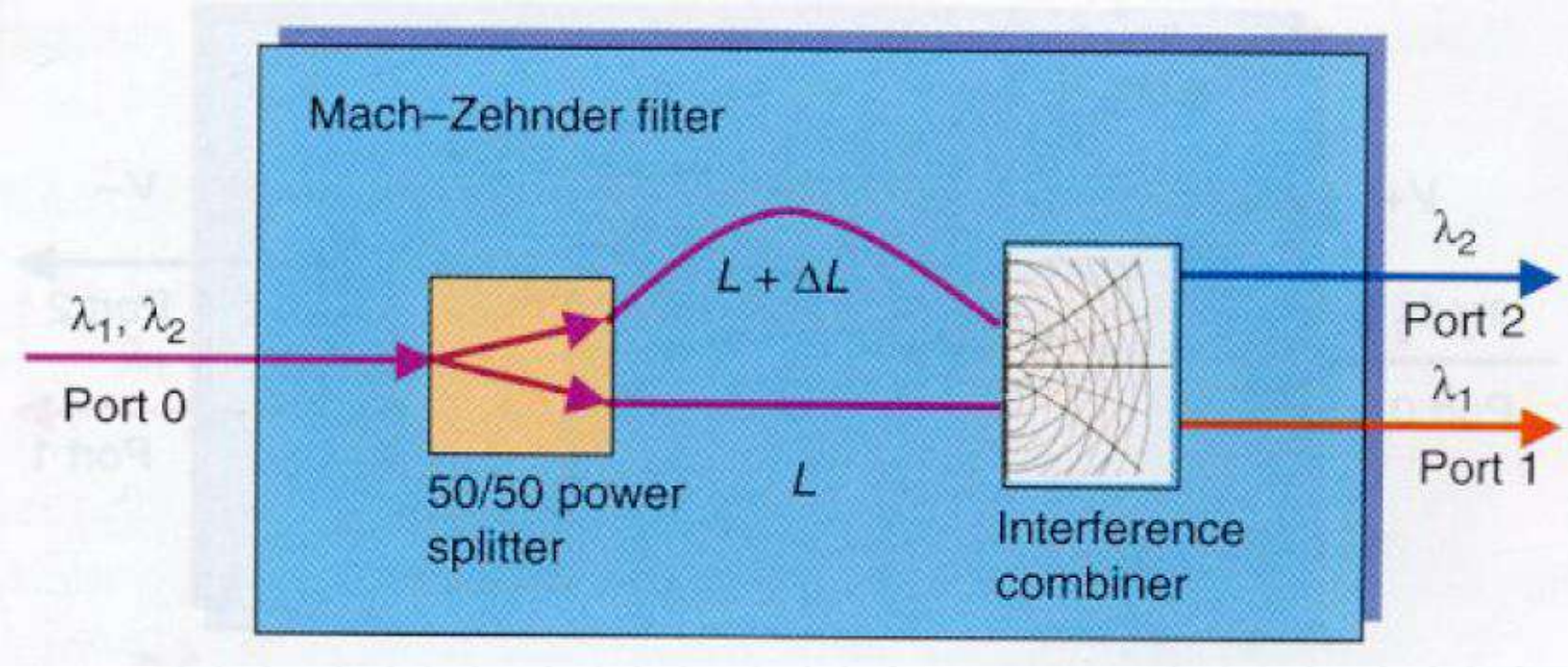

INPUT
REFLECTION
OUTPUT


I
λ
light spectrum
fibre core
fibre cladding
I
λ
transmission spectrum
Λ1
Λ2
Λ3
Bragg grating
I
λ
reflection spectrum

Mach-Zehnder Filter/Interferometer (MZI)

- In a Mach-Zehnder (MZ) interferometer, a splitter splits the incoming wave into two waveguides, and a combiner recombines the signals at the outputs of the waveguides
- An adjustable delay element controls the optical path length in one of the waveguides, resulting in a phase difference between the two signals when they are recombined.
- Wavelengths for which the phase difference is 180° are filtered out.
- By constructing a chain of these elements, a single desired optical wavelength can be selected.

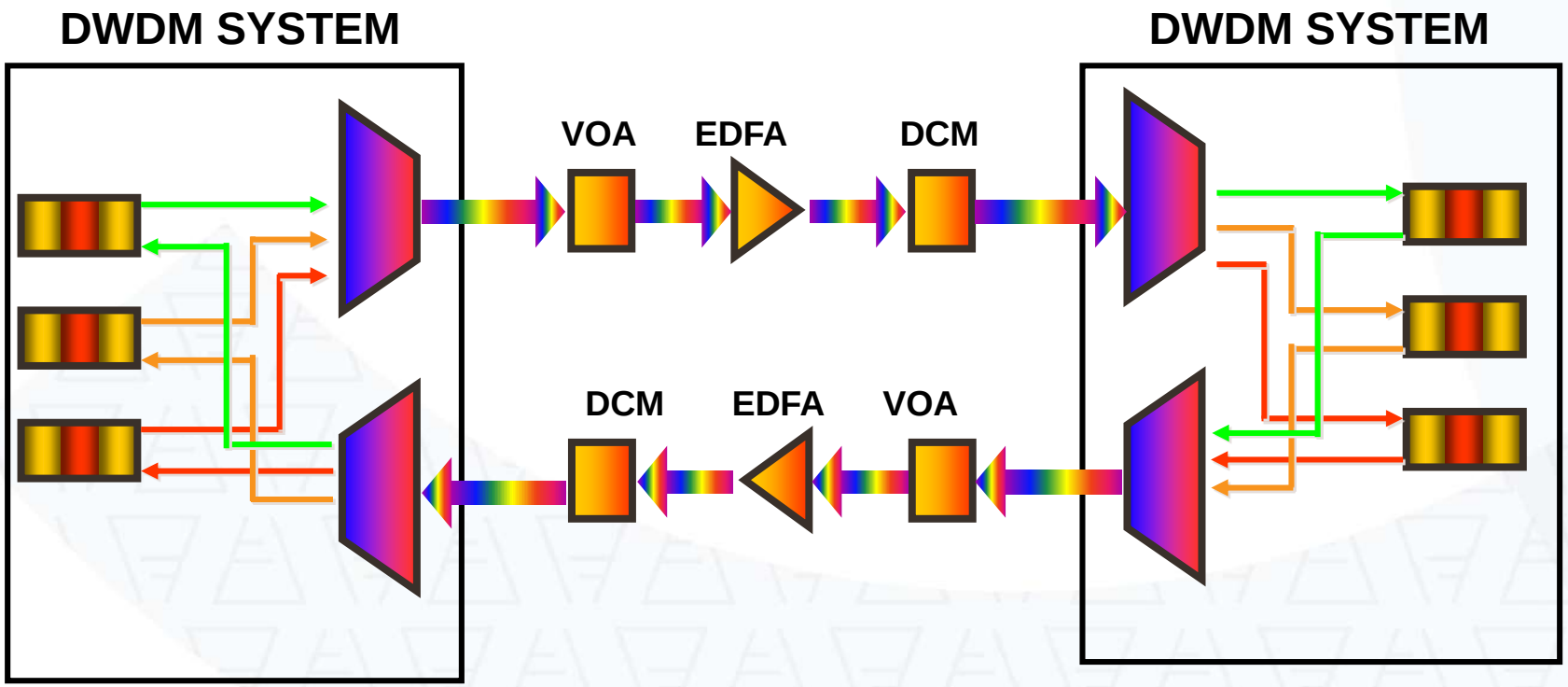
Mach-Zehnder Filter/Interferometer (MZI) *(cont.)*



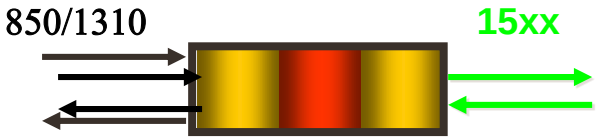
- Low-cost device
- Tunability: by varying temperature (~ few ms) [slow and complex]

WDM Network Technology

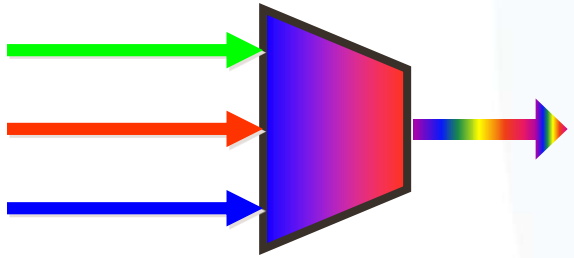
Typical DWDM Network



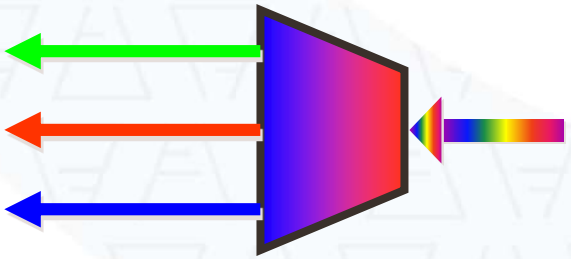
DWDM Components



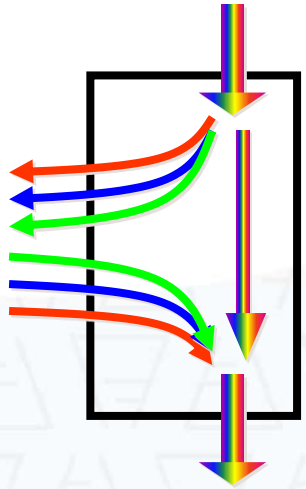
Transponder



Optical Multiplexer (MUX)

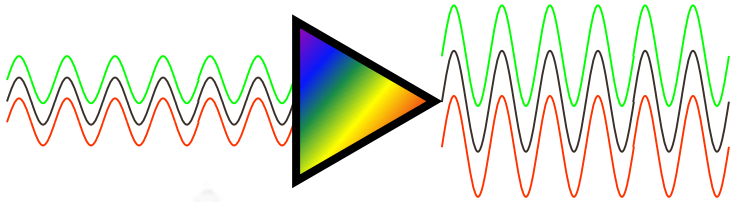


Optical De-multiplexer (DEMUX)

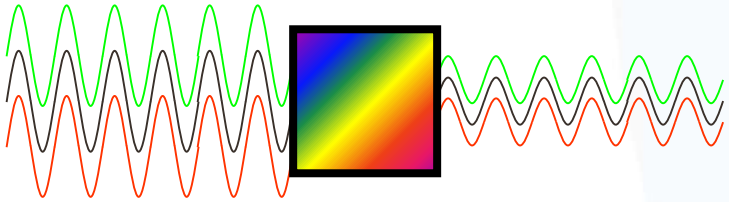


Optical Add/Drop Multiplexer (OADM)

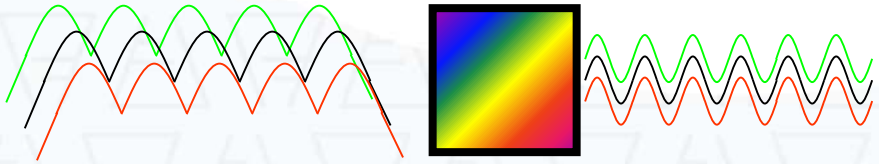
DWDM Components (Count.)



Optical Amplifier (EDFA)



Optical Attenuator
Variable Optical Attenuator



Dispersion Compensator (DCM / DCU)



THANK YOU

QUESTIONS?





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